

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 2 Summer 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 2

Subject Code & Name: 24AF1000ES206A - BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

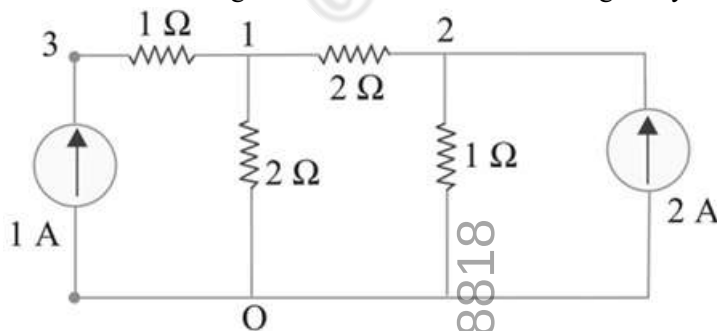
- 1 Kirchhoff's Current Law is based on the principle of:
 - a) conservation of voltage
 - b) conservation of power
 - c) conservation of energy
 - d) conservation of charge
- 2 The average value of a pure sinusoidal voltage over a full cycle is:
 - a) zero
 - b) peak value
 - c) RMS value
 - d) half of peak value
- 3 The working of an induction motor is based on:
 - a) mutual induction between stator and rotor
 - b) electrostatic interaction
 - c) back EMF generation
 - d) direct mechanical coupling
- 4 The working principle of a DC motor is based on:
 - a) Faraday's law of electromagnetic induction
 - b) Lenz's Law
 - c) Ampere's circuital law
 - d) Lorentz force on a current-carrying conductor
- 5 A basic DC power supply includes which of the following blocks?
 - a) Modulator, Mixer, Amplifier
 - b) Transformer, Rectifier, Filter, Regulator
 - c) Oscillator, Comparator, Detector
 - d) Transmitter, Load, Decoder
- 6 Zener diode acts as a voltage regulator by:
 - a) Storing energy in magnetic fields
 - b) Providing a constant current
 - c) Maintaining constant voltage despite input or load variations
 - d) Amplifying input voltage

- 7 The current gain in common-base (CB) configuration is:
 a) Less than 1
 b) Greater than 1
 c) Equal to 1
 d) Infinite
- 8 The output characteristic of a CE transistor has:
 a) A rising slope in saturation region
 b) A flat region in active mode
 c) A negative slope in cutoff
 d) An exponential rise
- 9 In a Digital Storage Oscilloscope (DSO), the analog signal is first converted into:
 a) Square wave
 b) Current signal
 c) Digital signal using ADC
 d) Frequency signal using FFT
- 10 A function generator can produce which of the following waveforms?
 a) Sine
 b) Square
 c) Triangle
 d) All of the above
- 11 In which region does a transistor operate when used as an amplifier?
 a) Cut-off
 b) Saturation
 c) Active
 d) Breakdown
- 12 A capacitor in a circuit:
 a) Blocks both AC and DC
 b) Passes both AC and DC
 c) Passes DC and blocks AC
 d) Passes AC and blocks DC

Q2. Solve the following.

- A) Determine the voltages 1 and 2 of the network in Figure by nodal analysis.

6



- B) State and explain Kirchhoff's Laws with examples.

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Q3. Solve the following.

- A) Describe the construction and working principle of a DC motor.
 B) Difference between Generator and Motor.

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Q4. Solve Any Two of the following.

- A) With the help of a block diagram, explain the working of a regulated DC power supply.
 B) Explain the working of a full-wave bridge rectifier with a neat diagram.
 C) Draw and explain the V-I (voltage-current) characteristics of a PN junction diode.

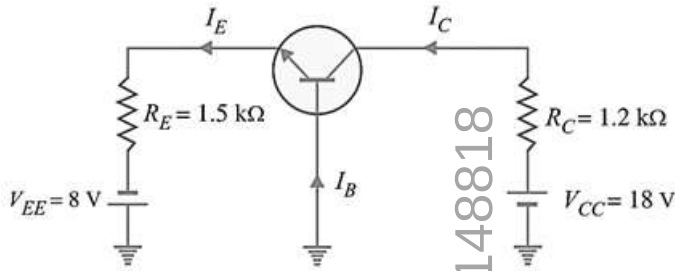
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Q5. Solve Any Two of the following.

- A) Describe the working of a PNP transistor with the help of a circuit diagram. 6
- B) With the help of a circuit diagram, explain the operation of a transistor as an amplifier. 6
- C) For the common base circuit shown in Figure, determine I_C and V_{CB} . Assume the transistor to be of silicon. 6



Q6. Solve Any Two of the following.

- A) Differentiate between Moving Coil and Moving Iron instruments. 6
- B) Draw and explain the block diagram of a Digital Storage Oscilloscope (DSO). 6
- C) Write a short note on a Function Generator. 6

*** End ***